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from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, confusion_matrix, classification_report,
    roc_auc_score, roc_curve
)
import matplotlib.pyplot as plt

# 2. Load Breast Cancer Dataset
data = load_breast_cancer()

X = data.data
y = data.target

# 3. Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

# 4. Feature Scaling
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# 5. Create SVM Model
svm = SVC(
    kernel='rbf',
    probability=True,
    random_state=42
)

# 6. Hyperparameter Tuning
parameters = {
    'C': [0.1, 1, 10, 100],
    'gamma': ['scale', 0.01, 0.001, 0.0001]
}

grid = GridSearchCV(
    svm,
    parameters,
    scoring='f1',
    cv=5
)

grid.fit(X_train, y_train)

print("Best Parameters:", grid.best_params_)
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# 7. Get Best SVM Model
model = grid.best_estimator_

# 8. Prediction
y_pred = model.predict(X_test)

# Probability for ROC-AUC
y_prob = model.predict_proba(X_test)[:, 1]

# 9. Calculate Metrics
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
auc = roc_auc_score(y_test, y_prob)
cm = confusion_matrix(y_test, y_pred)

# 10. Display Results
print("\n===== SVM RESULTS =====")

print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall   :", round(recall, 4))
print("F1-Score :", round(f1, 4))
print("ROC-AUC  :", round(auc, 4))

print("\nConfusion Matrix:")
print(cm)

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# 11. Plot ROC Curve
fpr, tpr, threshold = roc_curve(y_test, y_prob)

plt.plot(fpr, tpr, label="SVM AUC = " + str(round(auc, 3)))
plt.plot([0, 1], [0, 1], '--')

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - SVM")
plt.legend()
plt.grid()
plt.show()
```

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Best Parameters: {'C': 10, 'gamma': 0.01}
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==== SVM RESULTS ====
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Accuracy : 0.9825
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Precision: 0.9861
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Recall   : 0.9861
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F1-Score : 0.9861
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ROC-AUC  : 0.9977
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Confusion Matrix:
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```
[[41  1]
```

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 [ 1 71]]
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Classification Report:
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	precision	recall	f1-score	support
0	0.98	0.98	0.98	42
1	0.99	0.99	0.99	72
accuracy			0.98	114
macro avg	0.98	0.98	0.98	114
weighted avg	0.98	0.98	0.98	114

